

Christophe Hounwanou

AI Researcher - PhD Student in Computer Science at Université Laval
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RESEARCH INTERESTS

Artificial Intelligence, Reinforcement Learning, Machine Learning, Data Science.

EDUCATION

Université Laval, Quebec, Canada 2026 – Present
PhD in Computer Science – *Focus: Reinforcement Learning and Generative Models*

African Institute for Mathematical Sciences, Kigali, Rwanda 2024 — 2025
MSc in Mathematical Sciences: Data Science Grade: Very Good Pass
Thesis Title: Synthetic Financial Data Generation for Enhanced Financial Modeling

University of Abomey-Calavi, Abomey-Calavi, Benin 2020 — 2023
Bachelor's degree: Computer Science (Software Engineering) Cumulative GPA: 16.37/20
Thesis Title: Development of a Business Process Management Application at SBIN/DRSI/DDF

Faculty of Science and Technology, University of Abomey-Calavi, Benin 2020 — 2021
Bachelor's degree in Applied Mathematics and Computer Science Grade: Good

ACADEMIC EXPERIENCE

AIRINA Lab Remote / Canada
Junior AI Researcher July 2025 — Present

Etrilabs Cotonou, Benin
Data Scientist April 2024 — March 2025

Celtiis Benin / SBIN SA Cotonou St Michel, Benin
Data Analyst Intern April 2023 — December 2023

Collège Privé d'Enseignement General QUARTIER LATIN EM Abomey-Calavi, Benin
Part-time Mathematics Instructor October 2020 — November 2022

PUBLICATIONS & CONFERENCES

Preprints

- **Evaluating Generative Models for Synthetic Financial Data.**
Christophe D. Hounwanou (AIMS Rwanda), Ulrich Yae Gaba (AIMS RIC), Pierre Ntakirutimana (Carnegie Mellon University).
arXiv preprint arXiv:2512.21791, 2025.
- **Deep Generative Models for Synthetic Financial Data: Applications to Portfolio and Risk Modeling.**
Christophe D. Hounwanou (AIMS Rwanda), Ulrich Yae Gaba.
arXiv preprint arXiv:2512.21798, 2025.

Manuscripts Under Review

- **Deep Generative Models for Synthetic Financial Data: Applications to Portfolio and Risk Modeling.**
Under review at the *43rd International Conference on Machine Learning (ICML 2026)*.

Conference Presentations

- **SOSCHI Conference**, Kigali, Rwanda, 2025.
Synthetic Data Generation for Climate-Health Indicators: Leveraging AI to Overcome Data Gaps in Environmental Health Modelling.

AWARDS

Mastercard Foundation Funds for Alumni Start-ups in Transition (MCF FAST) Award	Rwanda
Youth Challenge Fund 3 - Plan International	Benin
Mastercard Foundation Graduate Scholarship	Rwanda
Mastercard Foundation Scholarship at the University of Abomey-Calavi	Benin

OTHER EXPERIENCES

Ministry of Digitalization and ICT of Benin	Ganhi, Benin
<i>Backend Developer Intern</i>	August 2022 — November 2022

SKILLS

- **Programming:** C/C++, Python, Tensorflow, Pytorch, Hugging Face
- **Software:** Git/GitHub, Docker.
- **Soft Skills:** Problem-solving, Teamwork, Communication, Adaptability, Leadership, Time management.

REFERENCES

Prof Aklilu Zeleke

Professor of Mathematics, Lyman Briggs College & Dept of Statistics and Probability, Michigan State University

E-mail: zeleke@msu.edu

Scholar Profiles: Michigan State University - Personal Pag — Google Scholar —

Prof. Eugène C. Ezin

Full Professor in Artificial Intelligence and Computer Science, Director of IFRI-UAC, University of Abomey-Calavi, Benin

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LinkedIn

Dr. Ulrich Yae Gaba

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